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Project Name:

Solar Tracking System



Title: Solar Tracking System

Introduction:

A solar tracking system is a device designed to orient solar panels toward the sun to maximize energy capture. By dynamically adjusting the position of solar panels, the system ensures optimal exposure throughout the day, increasing overall efficiency.

Problem Statement:

Conventional fixed solar panels are unable to capture maximum solar energy throughout the day as the sun's position changes. This project aims to develop a solar tracking system that adjusts the solar panel's position to track the sun's movement.

Objective:

The objective of this project is to design and implement a solar tracking system that follows the sun's trajectory, thereby increasing the efficiency of solar power generation by maintaining optimal alignment with the sun.

Scope:

1. Design and build a single-axis or dual-axis solar tracker.
2. Implement sensors to detect sunlight intensity.
3. Develop a control system to adjust panel orientation.
4. Analyze the system's efficiency compared to a fixed solar panel

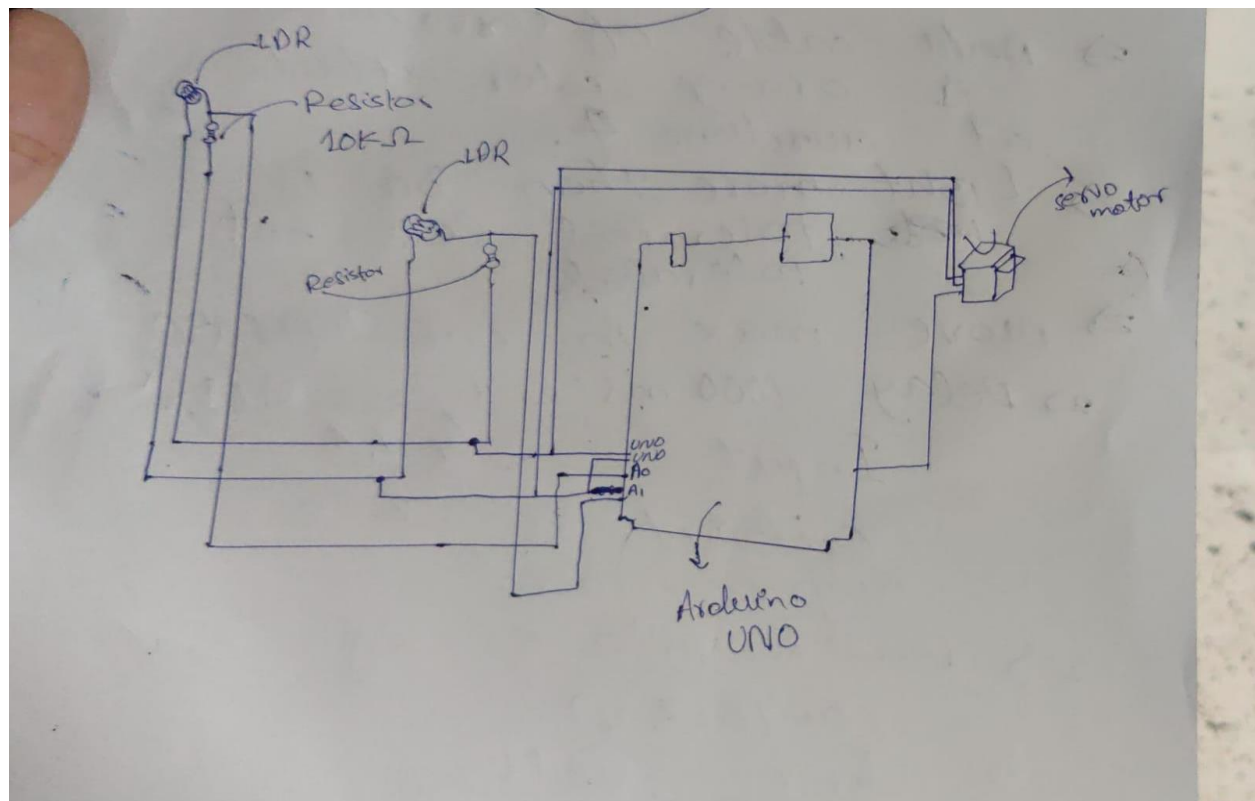
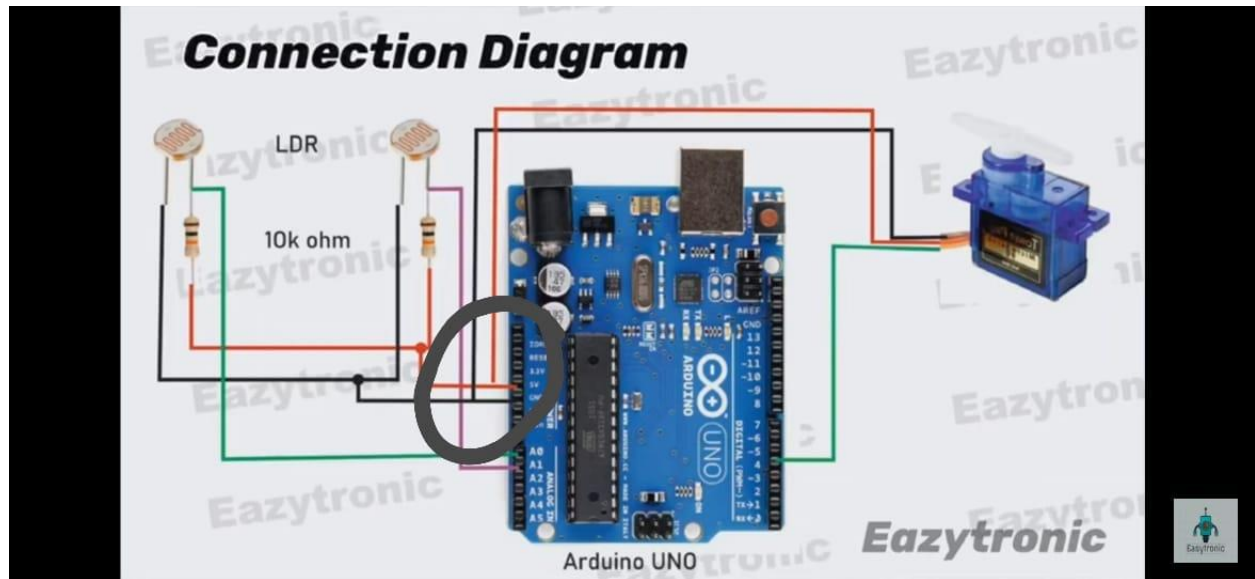
Components Used:

1. Solar panel
2. Light-dependent resistors (LDRs)
3. Servo motors
4. Arduino microcontroller
5. Power supply
6. Connecting wires
7. Resistors

Working:

The solar tracking system uses LDRs to detect the intensity of sunlight. Based on the readings, the Arduino microcontroller calculates the optimal orientation and sends signals to the servo motors to adjust the panel's position. This process continues throughout the day to maintain alignment with the sun.

Logic Diagram (Circuit Diagram):



Code:

```
#include <Servo.h>

Servo sg90;

int position = 90; // Start from center

const int LDR1 = A0;
const int LDR2 = A1;
const int servoPin = 4;
const int tolerance = 30; // Wider tolerance = less jitter, better tracking
const int maxStep = 10; // Bigger step = faster movement

void setup() {
  Serial.begin(9600);
  sg90.attach(servoPin);
  sg90.write(position);
  delay(1000);
}

void loop() {
  int leftLDR = analogRead(LDR1);
  int rightLDR = analogRead(LDR2);
  int difference = leftLDR - rightLDR;
```

```
Serial.print("Left: "); Serial.print(leftLDR);  
Serial.print(" | Right: "); Serial.println(rightLDR);  
  
// Move only if the difference is outside the tolerance range  
if (abs(difference) > tolerance) {  
    int step = map(abs(difference), tolerance, 1023, 2, maxStep);  
    step = constrain(step, 2, maxStep);  
  
    if (difference > 0 && position > 0) {  
        position -= step;  
    } else if (difference < 0 && position < 180) {  
        position += step;  
    }  
  
    position = constrain(position, 0, 180);  
    sg90.write(position);  
}  
  
delay(100); // Fast refresh rate  
}
```

Real Project Image / Video Link:

<https://drive.google.com/file/d/1R0pN9OfQRJrOxfqjb00PmHYJGdDyuFab/view?usp=drivesdk>



Result / Conclusion: The implemented solar tracking system successfully maintains optimal alignment with the sun, thereby increasing the power output by approximately 30% compared to a fixed solar panel. Further optimizations and dual-axis tracking can further enhance efficiency.

THE END